Lab Assignment: Case Study on Use Case Diagram

Case Study 1: Library Management System

Actors

Primary Actors

1. Member
2. Librarian
3. Administrator

Identified Use Cases

Member Use Cases

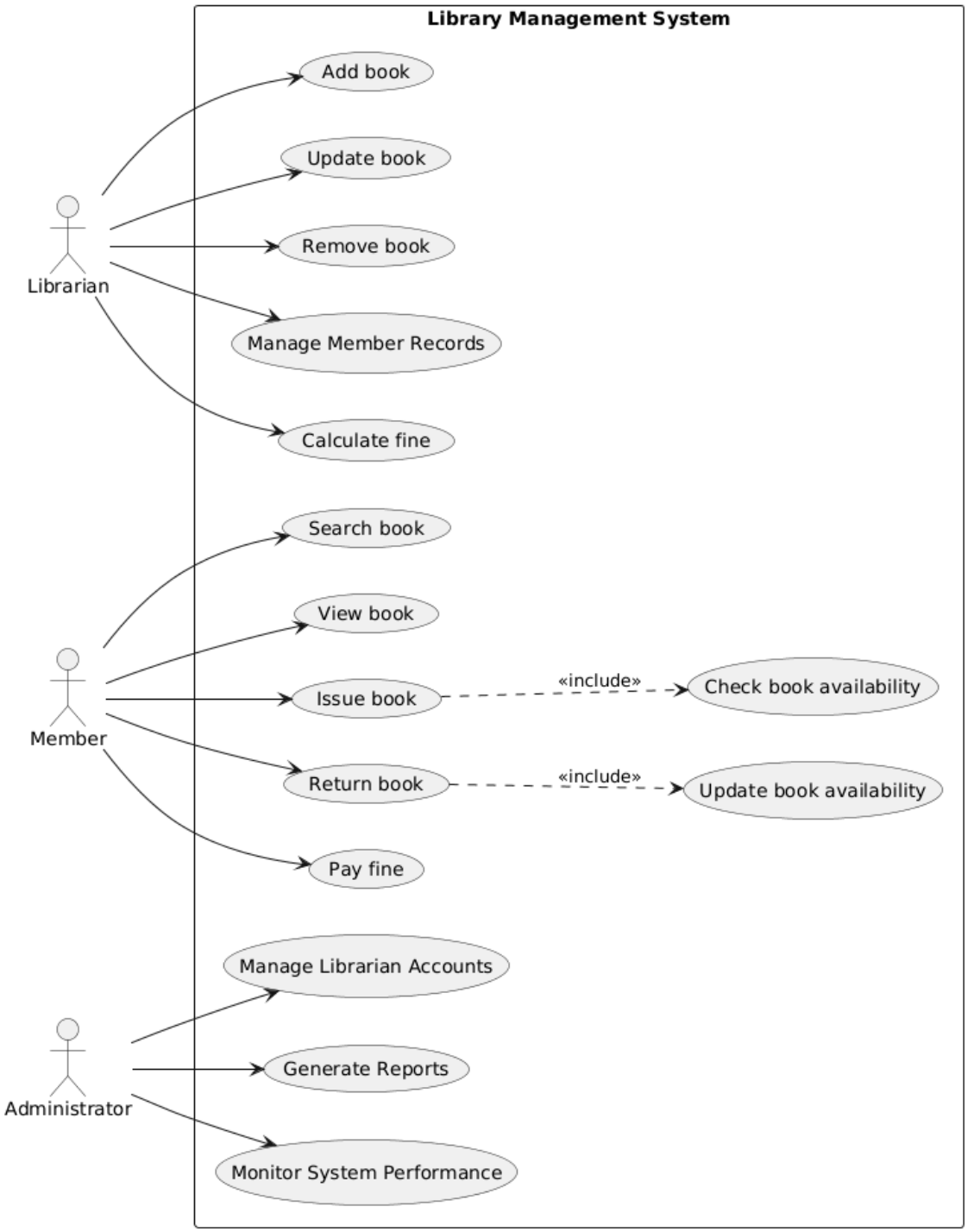
- Search Book
- View Book Details
- Issue Book
- Return Book
- View and Pay Fine

Librarian Use Cases

- Add Book
- Update Book
- Remove Book
- Manage Member Records
- Calculate Fine

Administrator Use Cases

- Manage Librarian Accounts
- Generate Reports
- Monitor System Performance



Case Study 2: Online Hospital Appointment & Care Management

1. Actors

Primary Actors

1. **Patient**
2. **Doctor**
3. **Lab Technician**
4. **Hospital Administrator**

External Actor

5. **Notification Service** (external system)

2. Identified Use Cases (Actor-wise)

Patient Use Cases

- Register Account
- Login
- Search Doctor
- View Doctor Details
- Book Appointment
- Reschedule Appointment
- Cancel Appointment
- View Appointment History
- View Prescriptions

- Download Medical Reports
- Raise Query / Complaint
- View Notifications

Doctor Use Cases

- Login
- Manage Schedule
- View Appointments
- Access Patient Medical History
- Update Diagnosis
- Add / Update Prescription

Lab Technician Use Cases

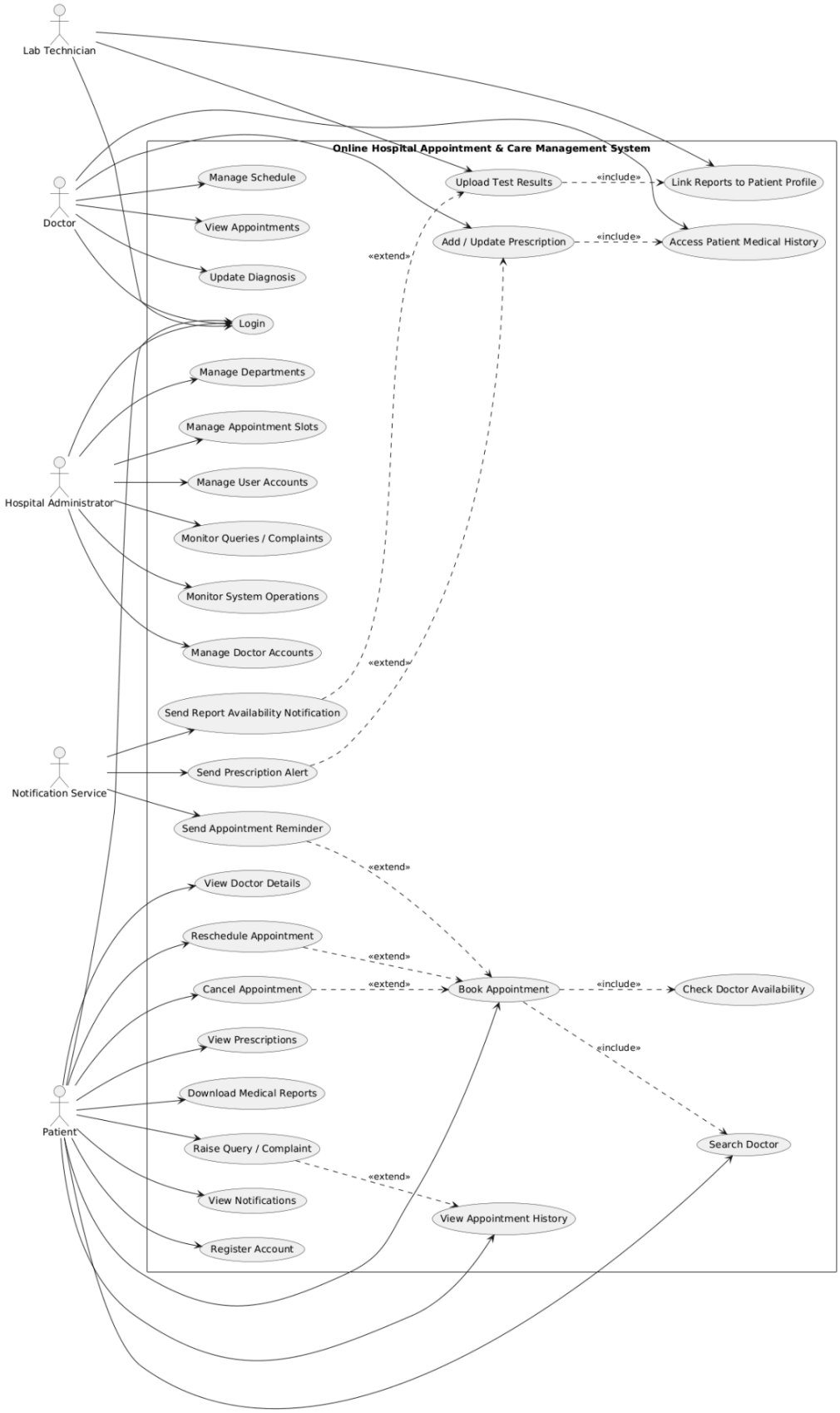
- Login
- Upload Test Results
- Link Reports to Patient Profile

Hospital Administrator Use Cases

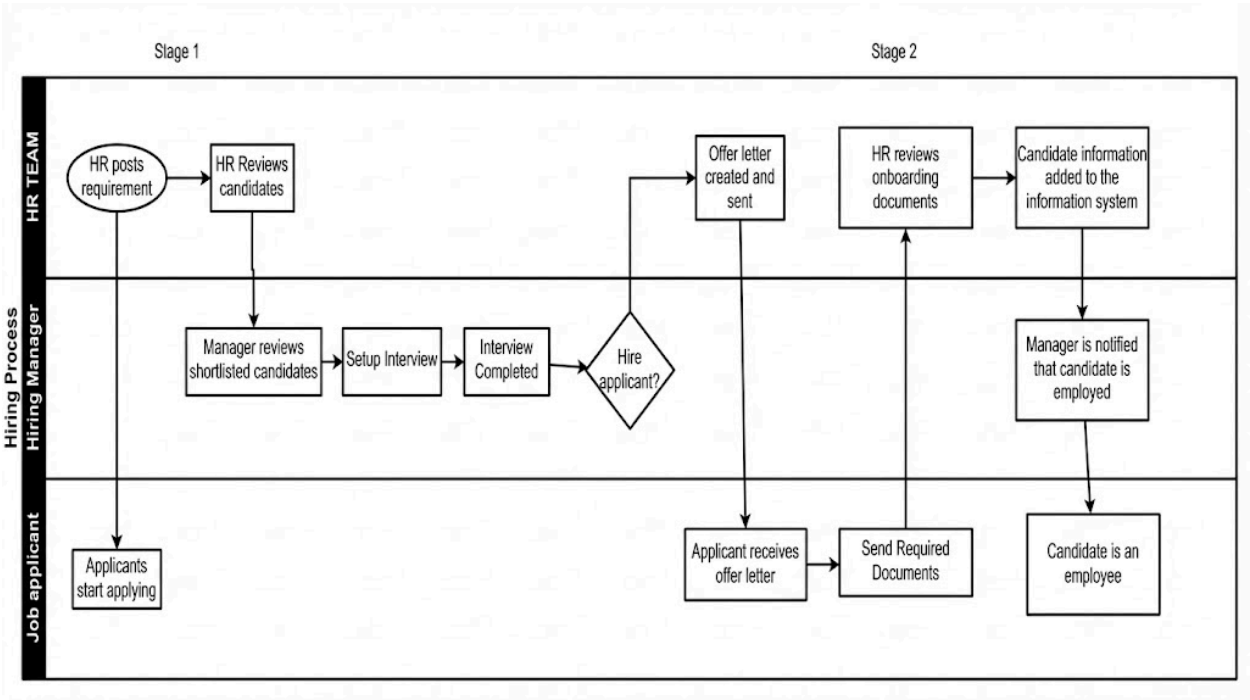
- Login
- Manage Doctor Accounts (Onboard / Update / Remove)
- Manage Departments
- Manage Appointment Slots
- Manage User Accounts
- Monitor Queries / Complaints
- Monitor System Operations

Notification Service Use Cases

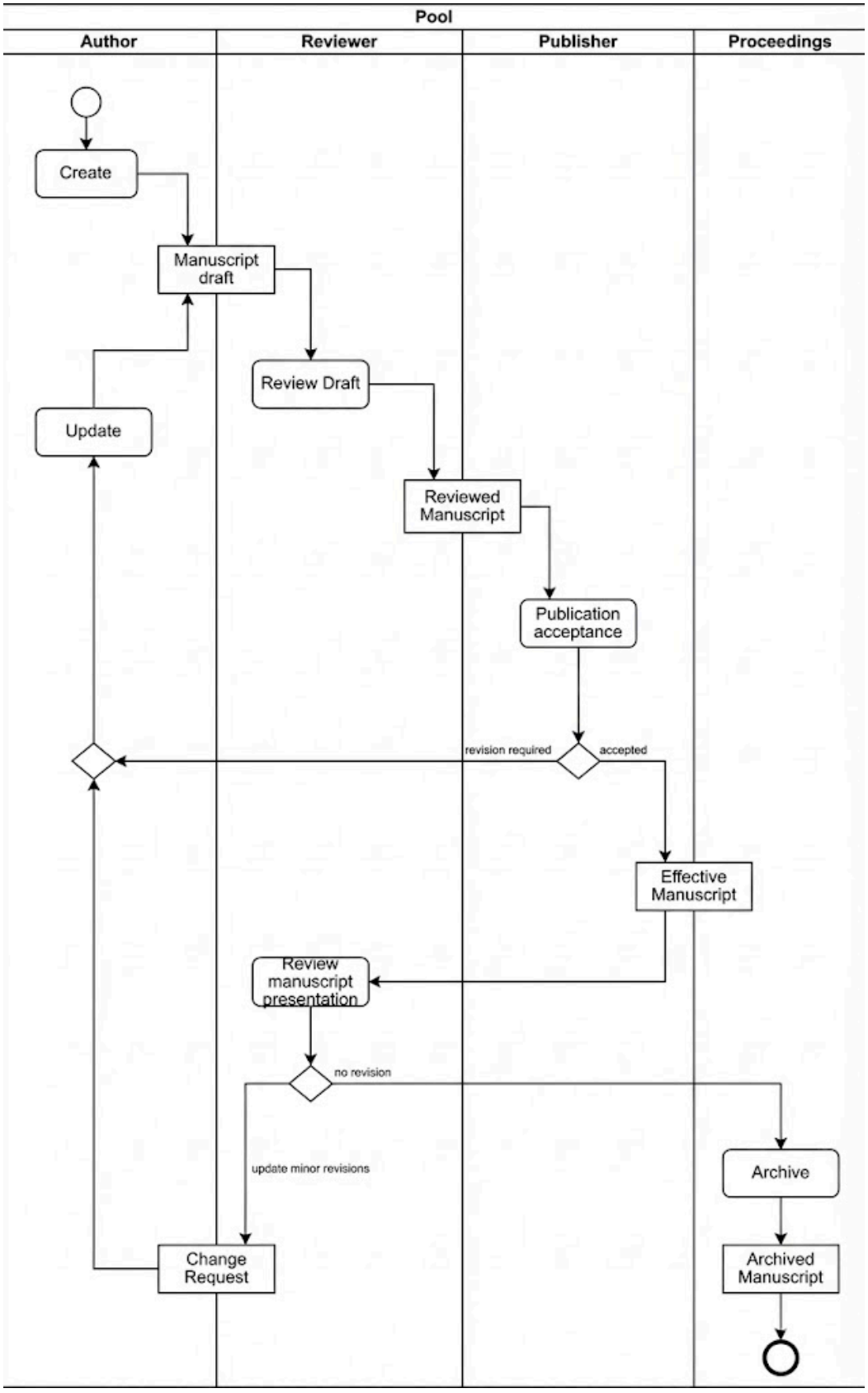
- Send Appointment Reminder
- Send Prescription Alert
- Send Report Availability Notification


Online Hospital Appointment & Care Management System
Lab Technician
Doctor
Hospital Administrator
Notification Service
Patient
Manage Schedule
View Appointments
Update Diagnosis
Upload Test Results
Add / Update Prescription
Login
Manage Departments
Manage Appointment Slots
Manage User Accounts
Monitor Queries / Complaints
Monitor System Operations
Manage Doctor Accounts
Send Report Availability Notification
Send Prescription Alert
Send Appointment Reminder
View Doctor Details
Reschedule Appointment
Cancel Appointment
View Prescriptions
Download Medical Reports
Raise Query / Complaint
View Notifications
Register Account
Link Reports to Patient Profile
Access Patient Medical History
Book Appointment
Check Doctor Availability
Search Doctor
View Appointment History
«include»
«extend»

Activity Diagram for Hiring Process



general process to publish a manuscript using


Pool
Author
Reviewer
Publisher
Proceedings
Create
Manuscript draft
Review Draft
Update
Reviewed Manuscript
Publication acceptance
revision required
accepted
Effective Manuscript
Review manuscript presentation
no revision
update minor revisions
Archive
Change Request
Archived Manuscript